

LOKPRAKASH ARAVINDAN

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PROFESSIONAL SUMMARY

Currently a Research Assistant in the Radiology Department at Stanford School of Medicine, with three years of specialized experience in cell culture, in vivo molecular imaging, and advanced biochemical - molecular and microbiological methodologies. Demonstrated proficiency in designing and executing intricate in vivo and in vitro experimental protocols, alongside quality control management. Recognized for exceptional organizational abilities and attention to detail, with a strong history of refining laboratory procedures and operations to maximize efficiency.

EDUCATION

University at Buffalo, The State University of New York | Buffalo, New York

Aug 2022 – Jan 2025

Master of Science: Chemical and Biological Engineering

B. S. Abdur Rahman Institute of Science and Technology | Tamil Nadu, India

Aug 2018 – Jul 2022

Bachelor of Technology: Biotechnology

SKILLS

- **Cell Culture & Quality Control:** In-vitro mammalian and stem cell culture (iPSC, hPSC), cryopreservation, cell inventory management, spheroid formation, cell counting, lab biosafety and phytochemical tests.
- **Biochemical & Molecular Techniques:** qRT-PCR, Western blot, Immunohistochemistry, ELISA, Microfluidics, gel electrophoresis, gene knockdown (siRNA), Plasmid transformation, MTT, CTB, flow cytometry (FACS), Immunostaining, Protein extraction, DNA/ RNA isolation, cDNA synthesis, media and buffer preparation.
- **Microscopy & Imaging:** In vivo Optical Imaging, Fluorescence and Confocal microscopy, Fluorescence imaging.
- **Lab Management:** Ordering lab supplies, Maintaining inventories, Biosafety, SOP writing, machine fixtures.
- **Equipment Handling:** Ultrasound, Incubators, Hemocytometer, Pipettes, Incucyte, Spectrophotometer, Nanodrop, iBlot, Centrifuges, ChemoDoc, Thermocycler, Fluorescence Imager, Autoclave, Cryotanks, BSL-2, Sonicator, High pressure Homogenizer and troubleshooting lab equipment.
- **Software Proficiency:** FlowJo, Graphpad PRISM, R, MATLAB, MacOS, Microsoft Word, Excel, MS Office, PowerPoint data entry, typing speed of 70WPM.
- **Additional Skills:** In-vivo models rodent (rat and mice), fish and shrimp handling – Necropsy, Dosing techniques, Tumor measurement, Tissue culture and collection, Advanced In vivo techniques - surgery and anesthesia, In vivo imaging.

PROFESSIONAL EXPERIENCE

Research Assistant

MIPS lab | Stanford University, CA

Feb 2025 – Present

- Investigated estrogen receptor activity and interactions to identify therapeutic strategies for triple-negative breast cancer.
- Developed mouse models to track B7-H3 activity in renal cancer, revealing its key role in suppressing T cell tumor attack and advancing immunotherapy strategies that boost T cell responses against cancer.
- Developed and executed comprehensive experimental protocols to investigate SUMOylation of Amyloid Precursor Protein (APP) in cellular models of Alzheimer's disease.
- Engineered novel microbubble-based contrast agents for enhanced ultrasound imaging of tumor microenvironments, optimizing chemical composition and acoustic properties to improve diagnostic sensitivity and specificity.
- Managed various cell cultures and performed key laboratory procedures, including autoclaving, media preparation, bioluminescence imaging, high-pressure homogenization, confocal microscopy, flow cytometry (FACS), and viability assays such as MTT and CTB.
- Assisted senior research scientists and post-doctoral researchers in their projects, actively contributed to seminar and lab meetings, presented innovative ideas, and provided critical insights to enhance ongoing research.
- Identified and troubleshoot experimental challenges by analyzing results, refining protocols, and implementing solutions to improve research outcomes.

Graduate Research Assistant

STEMMIL | University at Buffalo, NY

Aug 2022 – Jan 2025

- Maintained and expertly cultured mammalian cells and stem cells, incorporating gene knockdown by siRNA transfection for various detailed assays including qRT-PCR, Western Blot, Gel electrophoresis, and Immunofluorescence.
- Optimized lab protocols, significantly increasing overall productivity by 70% with minimal available resources.
- Managed lab orders, inventory, and supplier communication; trained and supervised a team of 4 graduate researchers.
- Troubleshoot and promptly resolved critical equipment issues; actively participated in CGTE Journal Club conferences.

Graduate Student Assistant

Campus Living | University at Buffalo, NY

May 2024 – Jan 2025

- Managed resident inquiries and concerns, enhancing customer satisfaction and fostering a supportive community environment in campus housing through proactive communication, efficient problem-solving, and exceptional service.

Graduate Intern | Levim Lifetech LLP, India**March 2019 – August 2019**

- Operated downstream production processes, including handling bioreactors, cooling equipment, HPLC, and autoclave parts.

Graduate Intern | Actavis Pharma Development Centre Pvt. Ltd., India**July 2019 – December 2019**

- Worked in microbiology lab; prepared media, calibrated equipment, trained in SOP documentation, drug analysis, and sorting.

Graduate Intern | SGS India Pvt. Ltd., India**April 2018 – September 2018**

- Interned in microbiology lab; trained in PPE, Gram staining, bacterial culture, hemocytometer use, and media preparation.

PROJECTS**Non-Genomic Estrogen Signaling Through ER36: Promotes Proliferation and Endocrine Resistance in TNBC****Stanford School of Medicine - Radiology****Nov 2025**

Investigated the activity and protein interactions of ER- α 36, a core Estrogen Receptor Isoform in aiding non genomic estrogen signaling inducing cell proliferation and implicated in breast cancer drug resistance. Developed and applied a split-luciferase bioluminescence complementation assay to measure ER- α 36 ligand-binding domain activity and its crosstalk with ER α and ER β in response to various ER related ligands. Complemented imaging data with cell proliferation assays, FACS-based live/dead and cell cycle analysis, and western blotting to characterize ER- α 36's role in non-genomic estrogen signaling and endocrine resistance.

SUMOylation of Amyloid Precursor Protein: A Steric Hindrance Strategy to Prevent Amyloidogenic Processing in Alzheimer's Disease | Stanford School of Medicine - Radiology**Feb 2025**

Investigated SUMOylation of Amyloid Precursor Protein (APP) as a strategy to sterically hinder secretase access, preventing amyloid- β (A β) production. This approach disrupts the amyloidogenic cascade at its origin, potentially reducing neurotoxic plaque formation and tau pathology, and highlights a promising therapeutic avenue for modifying Alzheimer's disease progression at the molecular level.

Early Diagnosis of Renal Cell Carcinoma using B7-H3 Targeted Molecular Imaging Probes**Stanford School of Medicine – Radiology****June 2025**

Established a preclinical model by injecting renal carcinoma (RENCA) cells into mice, enabling in vivo monitoring of B7-H3 expression dynamics in tumor development. Characterized B7-H3 activity to elucidate its immune checkpoint function and association with T cell-mediated tumor cytotoxicity. Demonstrated the approach's potential to uncover novel mechanisms for enhancing T cell responses and advancing targeted immunotherapies in renal cancer.

Partial Reprogramming of the Liver Gene Regulatory Network (GRN) in Hepatic Cells by Targeting HNF4a and FOXA1/2/3**University at Buffalo Dr. Natesh Parashurama's STEMMIL****Aug 2022 – Jan 2025**

Enhanced liver disease treatments by refining GRN involved in liver cell development using hPSCs, focusing on HNF4a and FOXA1/2/3. Utilized siRNA in HEPG2 cells; assessed results using qRT-PCR, Western Blot, Immunostaining, and fluorescence microscopy. Improved differentiation, maturity, and yield of hPSC-derived hepatocytes, providing insights for cell therapies.

Antidiabetic Potential of *Ipomea batatas* using Danio rerio | Dynamic Megaceutics | Dec 2021 – May 2022

Tested for the phytochemical properties of *Ipomea batatas* extract on control and diabetes-induced fish. Formulated a new diet to check for the anti-diabetic effects on fish. Dissected, performed assays and recorded results, watched for morphological changes.

Probiotic Supplements for *Pennaeus vannamei* Shrimp Species | Geomarine Biotechnologies | Aug 2021 – Oct 2021

Cultivated *Saccharomyces cerevisiae* yeast strains and meticulously extracted beta-glucans and oligo mannans for innovative shrimp supplements. Rigorously tested various concentrations on PL 20 for significant morphological changes and closely monitored growth.

Comparative Evaluation of Natural Plant Drugs for In-Vitro Antidiabetic Activity | Avigen Biotech | Jul 2021 - Sep 2021

Plant and bark extraction which were then used to conduct comprehensive quantitative, qualitative, and phytochemical screening analysis on *Alternanthera sesilis* and *Cinnamomum verum* to rigorously evaluate their potent anti-diabetic activity.

CERTIFICATION AND OTHER DEGREES

Rodent Handling	UB LAF – Laboratory Animal Facility	Nov 2024
Basic Injections	UB LAF – Laboratory Animal Facility	Nov 2024
Mouse and Rat Advanced techniques	UB LAF – Laboratory Animal Facility	Nov 2024
Rodent Anatomy and Necropsy	UB LAF – Laboratory Animal Facility	Nov 2024
CRISPR in Gene Editing, Gene Regulation, ChIP, Live-Cell Imaging, Diagnosis	UDEMY (LINK)	Jun 2024
Human Stem cell oversight	Collaborative Institutional Training Initiative (LINK)	Jul 2024
Good Clinical Practice	Collaborative Institutional Training Initiative (LINK)	Jul 2024
Aseptic surgery – Lab animal research	Collaborative Institutional Training Initiative (LINK)	Jul 2024
Antibody production in animals	Collaborative Institutional Training Initiative (LINK)	Jul 2024
Working with Mice	Collaborative Institutional Training Initiative (LINK)	Jul 2024
Doctorate in Medicine in Acupuncture - M.D Acu.	Medicina Alternativa Alma Ata	Nov 2018
Laboratory Bio-safety Training	Environment health and safety (LINK)	Sep 2023

PUBLICATIONS

- In Vitro Cultivation Techniques for Modeling Liver Organogenesis, Building Assembloids, and Designing Synthetic Tissues using Human Cell Lines. Simran Kumar, Jenna Venturo, Helly Patel, Lokprakash Aravindan, Kyle Leatt, Natesh Parashurama

ACHIEVEMENTS

Awarded second place for the “**Early Diagnosis of Renal Cell Carcinoma using B7-H3 Targeted Molecular Imaging Probes**” research poster at the Canary Center Symposium "The Science of Seeing Disease Earlier" hosted by Stanford Radiology Department at Stanford University School of Medicine - April 2026